Concept & Outline

Scalar Formulation

$$
\begin{aligned}
f_1(x_1, x_2, \dots, x_n) &= 0 \\
f_2(x_1, x_2, \dots, x_n) &= 0 \\
&\vdots \\
f_n(x_1, x_2, \dots, x_n) &= 0
\end{aligned}
$$

Vector Formulation

$$\mathbf{f}(\mathbf{x}) = \mathbf{0}$$

$$\mathbf{f} = \begin{Bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{Bmatrix}, \mathbf{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}, \mathbf{0} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{Bmatrix}$$

Linear System:

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n
\end{aligned}
$$

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

= Coefficient Matrix

$\mathbf{b}$ = Constant Vector

There is no product of x's. All x's have power of one.

Nonlinear System: The system that is not linear.

③ Roots of Systems of Equations

Linear Solvers:

- Gaussian Elimination Method
  Obsolete method.
- Matrix Inversion Method
  The method that should be avoided.
- ✓ LU Decomposition Method
  Classical method appears in exam.
- Jacobi Iterative Method
  Diagonally dominant matrix.
- ✓ Gauss-Seidel Iterative Method
  Diagonally dominant matrix, popular method for exam.
- Successive Relaxation Method
  Diagonally dominant matrix.
- Gradient Descent Method
  Simple solver with slow convergence.
- ✓ Conjugate Gradient Method
  Basic solver for practical uses.

Nonlinear Solvers (Linearized system is solved and solution is updated reiteratively.):

- ✓ Newton-Raphson Method
  Fast convergence using exact Jacobian matrix.
- Broyden's Method
  Generalized secant method using approximated Jacobian matrix.